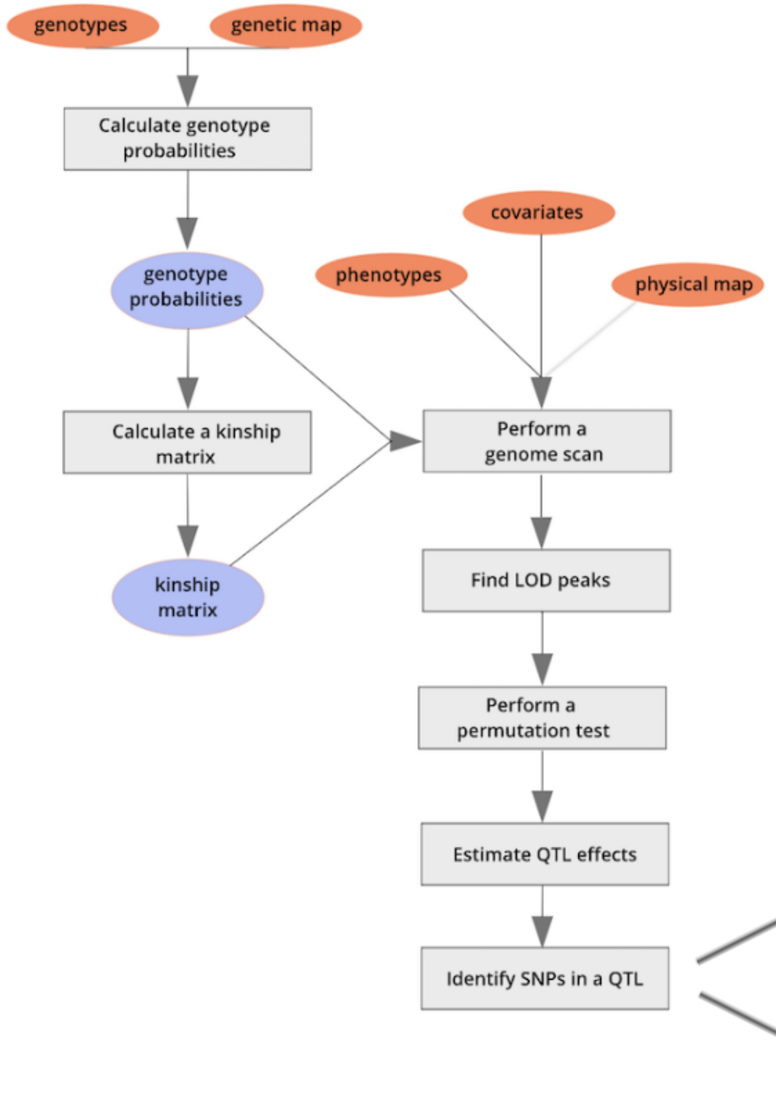
Pheno2GeneRank

Introduction: Biological Background of GWAS or QTL



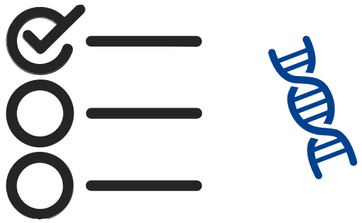
"Goal: To prioritize causal genes within QTL that is driving certain trait / phenotype"

Trait / Phenotype





Candidate Gene List





Gene Exploration & Selection

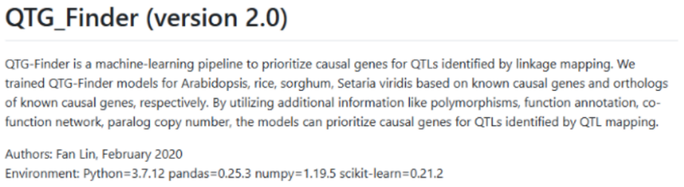
Previous tools (QTG_Finder, QTG-LGBM)

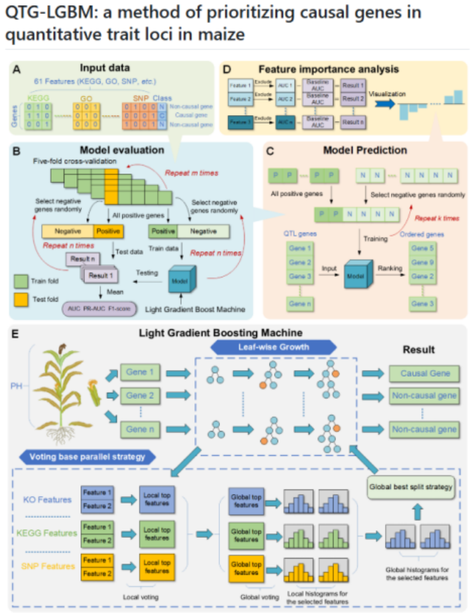
Feature Engineering
→ Machine Learning

Use only predefined features

Low Explainability

Do not use
trait / phenotype context

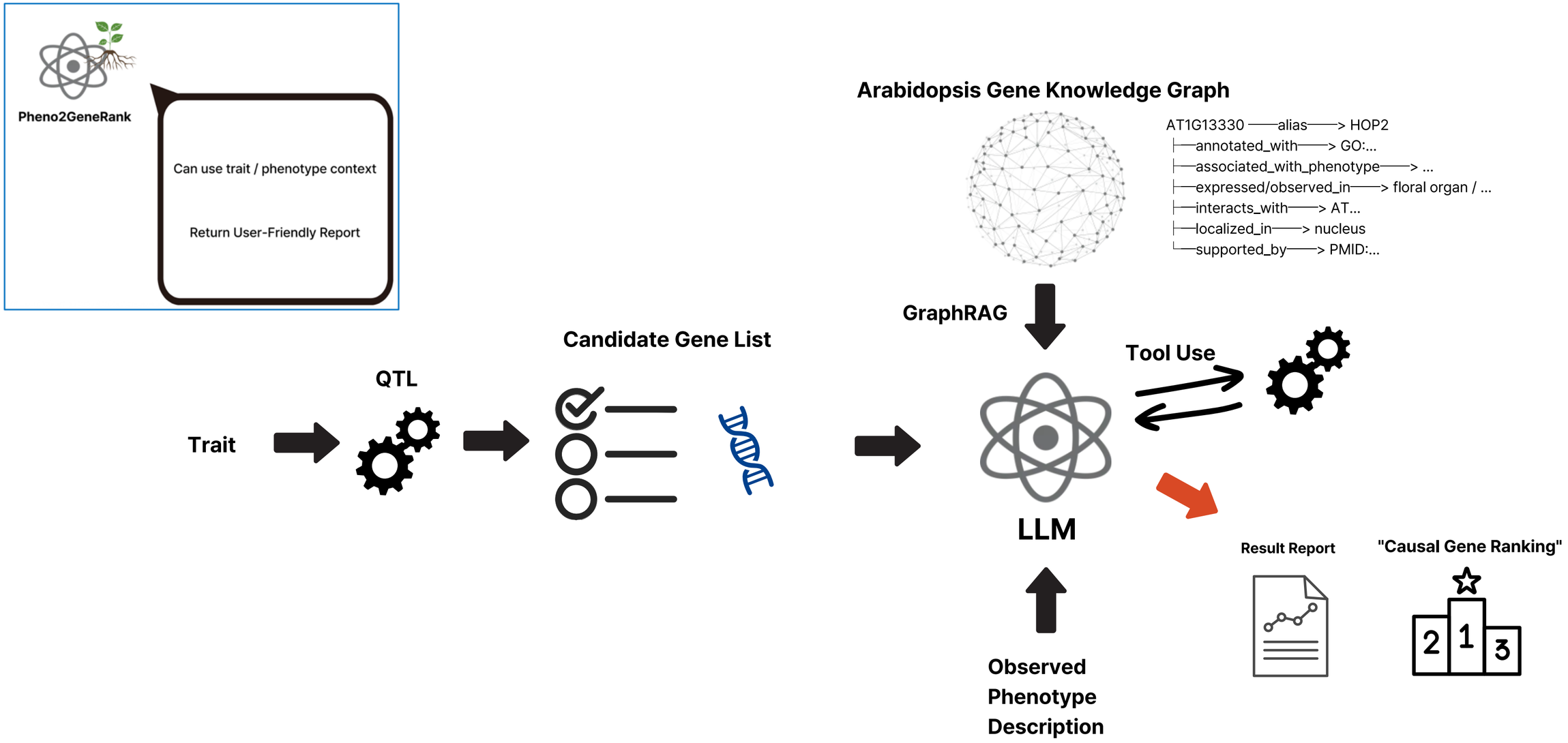




QTG-Finder : employs random forest (RF) to predict causal genes within flowering time (FT)-related QTL in Arabidopsis thaliana (Arabidopsis), *Oryza sativa* (rice), Setaria *viridis* (Setaria), and *Sorghum bicolor* (sorghum)

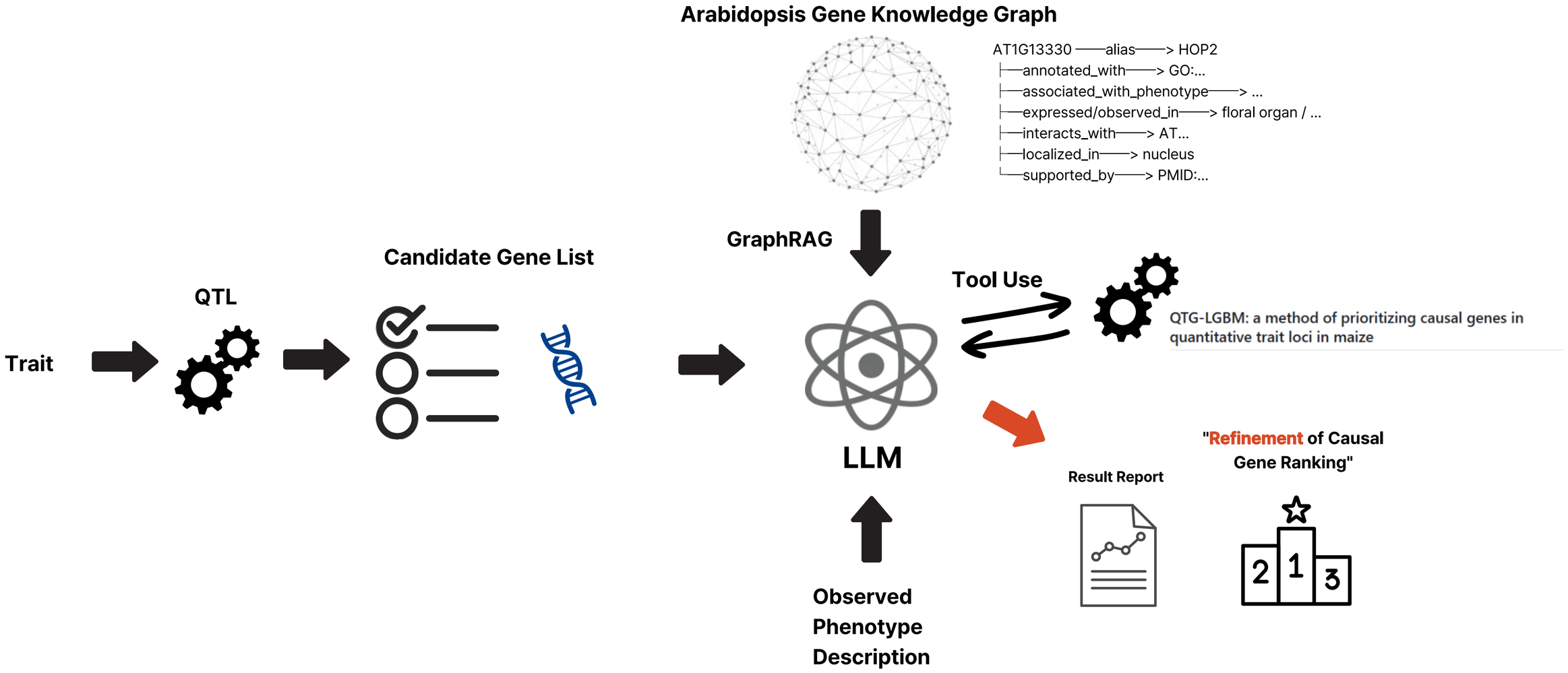
The input for ***QTG-LGBM*** involves 61 features about 25,629 genes (Fig. 1A). These gene features comprise 17 Kyoto Encyclopedia of Genes and Genomes (KEGG) features, 17 Gene Ontology (GO) features, and 27 Single Nucleotide Polymorphism (SNP) features.

Idea of Pheno2GeneRank: Causal Gene Ranking LLM Agent through Observed Phenotype Descriptions

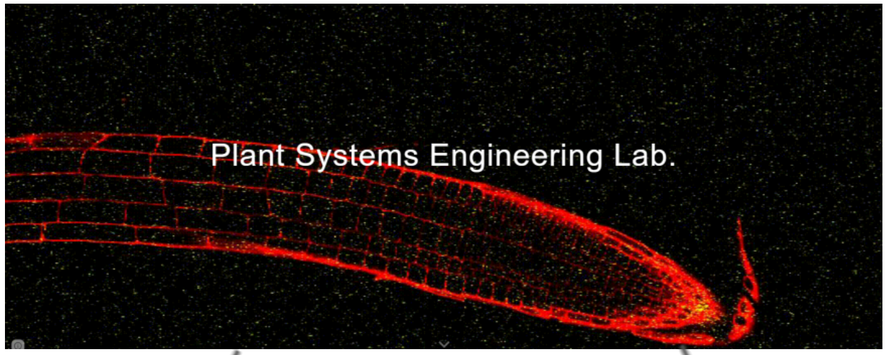


Idea 2 of Pheno2GeneRank: Use QTG-LGBM as a tool for Pheno2GeneRank

→ We can use both the predefined features (that are used for QTG-LGBM) and phenotype context



"Cowork"



Supervision
for Pheno2GeneRank

Plant System Research with Pheno2GeneRank



Systems Biology and Bioinformatics Laboratory

Pheno2GeneRank Development

1. Algorithm Design
2. Ablation Study

Evaluation

1. Metric Design
2. Test Dataset Construction
3. Validation

Step-by-step workflow

- Step1. Collect input data for testing
  - Which trait will be collected?
  - What kind of gold standard (known causal genes from QTL experiments) are available?

- Step2. Build LLM Agent
  - What kind of LLM agents will be constructed (idea 1 vs. idea 2)?
  - What will be the output (what will provide in the report)?
  - What will be the evaluation metric?

- Step3. Agent interpretation / Feature importance analysis / Experimental validation
  - Can we provide the reason why certain gene is causal?
  - How can we experimentally validate causal gene?

QTG-LGBM Operation

"Training & Validation & Test Data"

Common feature value in the same species: We can easily get the data from QTG-Finder

Curated Labels

ID	carb	nucleotide	energy	fatty/lipid	specialized		class
AT1G01010	0	0	0	0	0		0
AT1G01020	0	0	0	0	0		0
AT1G01030	0	0	0	0	0	....	0
AT1G01040	0	0	0	0	0		0
AT1G01050	1	1	0	0	0		0

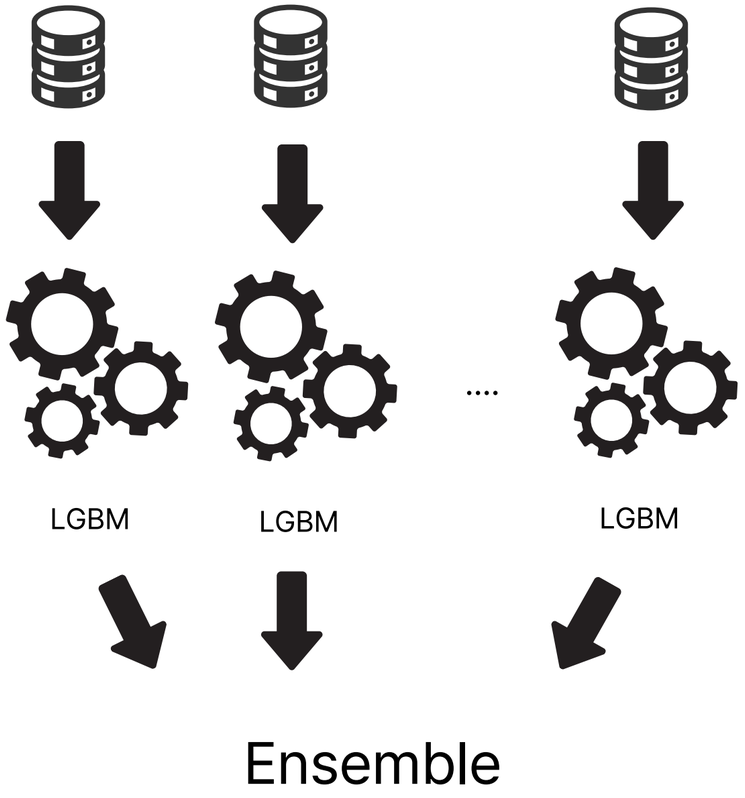
Some genes are used for training
Some genes are used for validation
The others are used for testing

We should do labeling again
when we change the focus from one trait to other

1: Ground-Truth Causal Gene
0: Unknown Gene (This can be a causal gene)

QTG-LGBM Operation

"Training & Inference"



All samples that have class 1 are used
Some samples that have class 0 are sampled

Training

Inference

QTG-LGBM Operation

"User Inference"

QTL Genes →  → Each gene is classified whether it is causal gene or not

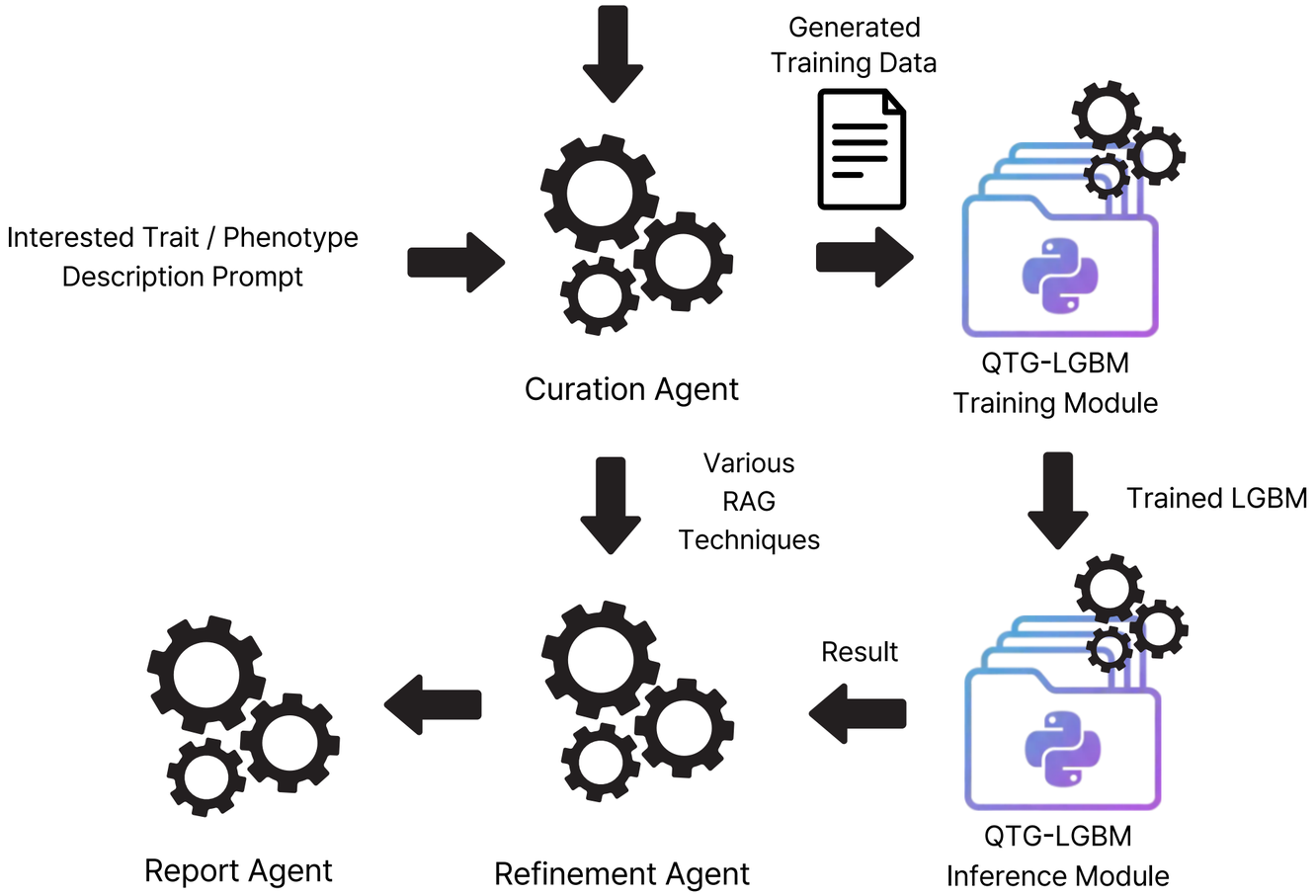
Trained QTG-LGBM

Classification probability is used for gene ranking

Multi-Agent Pipeline

"User Input File"

chr	peak_position	start	end	LOD	trait
Chr1	18,420,000	16,900,000	20,100,000	7.81	root_length
Chr3	7,830,000	6,950,000	8,620,000	5.32	root_angle



Curation Agent

1. Extract trait / phenotype word
2. Search QTL genes from the user input file
3. Labeling 1 / 0 class from external information source
4. Search the useful information used for a refinement agent

Refinement Agent

1. Refine the gene rank from QTG-LGBM
2. Refine only some class 0 genes using various sources

Report Agent

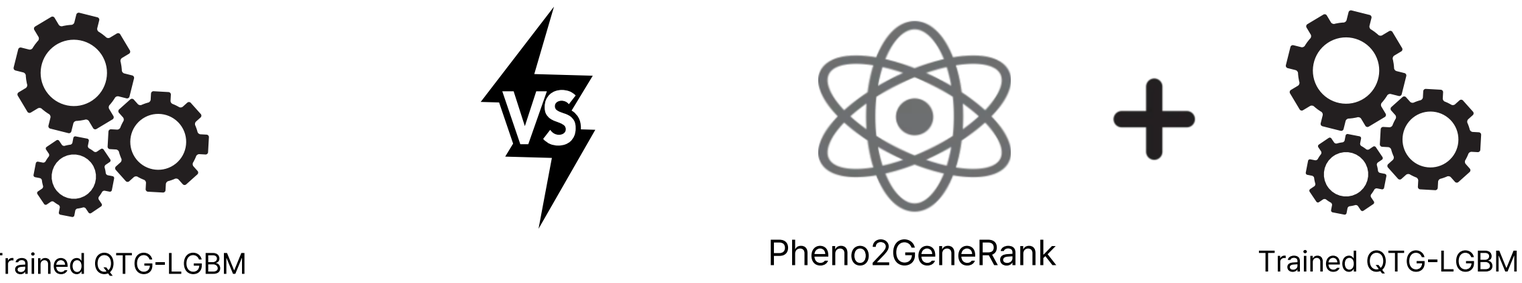
1. Summarize the results and return an user-friendly report file

Evaluation

Generated
Training Data

Some class 1 samples are used as Test Data

Must be same



Class 1 sample is included in the results or not